

Digital Logic and Computer Design (M. Morris Mano)

Chapter 1: Binary Systems

1.1 Digital Computers and Digital Systems

- The most striking property of a digital computer is its generality. It can follow a sequence of instructions, called a program, that operates on given data.
- The user can specify and change programs and/or data according to the specific need. As a result of this flexibility, general-purpose digital computers can perform a wide variety of information-processing tasks.
- Characteristic of a digital system is its manipulation of discrete elements of information. Such discrete elements may be electric impulses, the decimal digits, the letters of an alphabet, arithmetic operations, punctuation marks, or any other set of meaningful symbols.
- The juxtaposition of discrete elements of information represents a quantity of information.
- Discrete elements of information are represented in a digital system by physical quantities called signals. Electric signals such as voltages and currents are the most common.
- The signals in all present-day electronic digital systems have only two discrete values and are said to be binary.

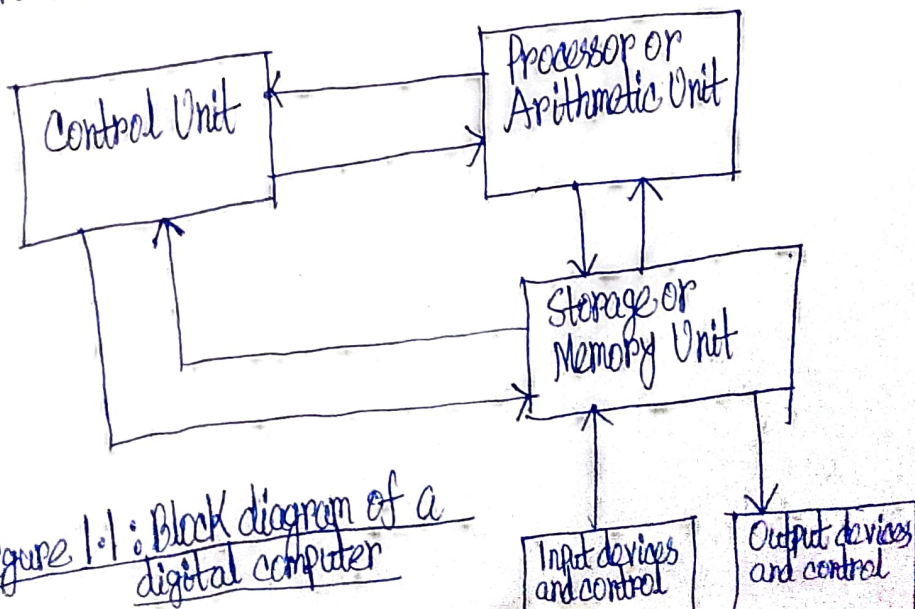


Figure 1.1: Block diagram of a digital computer